Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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This paper represents my own work in accordance with University regulations.
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Abstract

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1 Introduction

Large-scale density fluctuations contain fundamental information about the structure of many-body systems. In particular, characterizing the local number variance of point patterns has widespread applications across condensed matter physics and beyond. From quasicrystals that exhibit sharp diffraction with forbidden rotational symmetries [1] to the photoreceptor mosaics of bird retinas and the distribution of prime numbers on the real line [2], the large-scale density fluctuations are anomalously suppressed below what simple randomness would predict. The property that unites these systems is hyperuniformity.

For any given point pattern, we can consider the variance in the number of points in some arbitrarily-centered local observation window. For a one-dimensional random (Poisson) distribution, this variance grows linearly in proportion to the region's size, according to $\sigma^2(R) = 2\rho R$. This linear growth is characteristic of uncorrelated matter and is known as the Poisson baseline [3]. In a hyperuniform system, the variance will instead grow more slowly than the window size, even though the pattern may appear equally disordered at short distances [3, 2]. This can be formalized by the introduction of a structure factor $S(k)$, which measures the intensity of density fluctuations at wavenumber k. A point pattern is defined as hyperuniform if and only if $S(k) \to 0$ as $k \to 0$ [3].

Not all hyperuniform patterns suppress fluctuations equally. While for all hyperuniform patterns $S(k) \to 0$ as $k \to 0$, the rate at which $S(k)$ vanishes can vary. Generally, hyperuniform patterns are sorted into three main classes depending on how quickly $S(k) \to 0$ as $k \to 0$, assuming $S(k) \sim |k|^\alpha$. Here, α determines whether the number variance is bounded (Class I, $\alpha > 1$), grows logarithmically (Class II, $\alpha = 1$), or grows as a weak power law (Class III, $0 < \alpha < 1$) [2]. While this classification scheme is useful, it is still not able to fully distinguish between hyperuniform patterns.

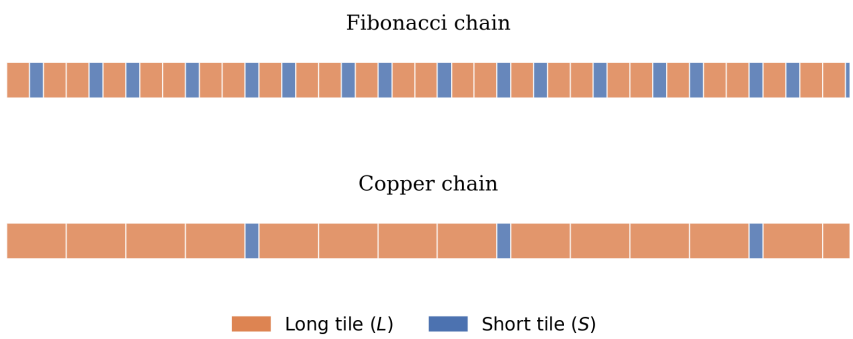

Figure 1: Two Class I hyperuniform chains, both with $\alpha = 3$, but with structurally distinct tile arrangements.

To distinguish between patterns within the same class, we need a metric that captures the amplitude of the fluctuations, not just their scaling rate. The surface-area coefficient $\bar{\Lambda}$, defined as the running average of the number variance at large R [3], serves this role. So far, $\bar{\Lambda}$ has only been computed for the integer lattice [3] and Fibonacci chain [4], leaving open the question of how other aperiodic hyperuniform patterns compare.

In this work, we compute both α and $\bar{\Lambda}$ for a broad collection of one-dimensional hyperuniform point patterns, including substitution tilings, perturbed lattices, and stealthy configurations. We identify the algebraic constraints that determine which α values are accessible, and we use $\bar{\Lambda}$ to rank patterns within each class.

2 Theory

We first define the number variance, structure factor, and the three hyperuniformity classes, then introduce the surface-area coefficient $\bar{\Lambda}$ as a finer ranking metric. We then describe the three families of patterns studied in this work: substitution tilings, perturbed lattices, and stealthy configurations.

2.1 Number variance and structure factor

Let us start by considering a one-dimensional point pattern with number density ρ. If we slide a window with half-width R along the pattern and count the number of points inside, the count will fluctuate from one window placement to the next. The variance of this count is the number variance, $\sigma^2(R)$, and how it grows with R tells us a great deal about the spatial structure of the pattern.

For a Poisson process, each part of the system is statistically independent of the rest, so bigger windows accumulate more noise and the variance grows linearly ($\sigma^2_{\text{Poisson}}(R) = 2\rho R$) [3]. When points are spatially correlated, however, the variance picks up a correction that depends on the pair correlation function $g_2(r)$, defined as the ratio of the two-particle density to ρ^2 [3]. When $g_2(r) > 1$, points are more likely to be found at separation r than in an uncorrelated system; when $g_2(r) < 1$, they are less likely. The variance is [3]:

$$\sigma^2(R) = 2\rho R \left[1 + 2\rho \int_0^{2R} \left(1 - \frac{r}{2R}\right) \left[g_2(r) - 1\right] dr \right]. \tag{1}$$

To connect the variance directly to the small-k behavior of the pattern, it is useful to work in Fourier space with the structure factor [3]

$$S(k) = 1 + \rho \hat{h}(k), \tag{2}$$

where $\hat{h}(k)$ is the Fourier transform of the total correlation function $h(r) = g_2(r) - 1$. The structure factor is non-negative for all k and equals 1 everywhere for a Poisson process. The number variance can then be rewritten as a weighted integral over $S(k)$ [3, 4]:

$$\sigma^2(R) = \frac{\rho}{\pi} \int_0^\infty S(k) \, \frac{4\sin^2(kR)}{k^2} \, dk. \tag{3}$$

Because $4\sin^2(kR)/k^2$ is peaked near $k = 0$ for large R, the large-scale behavior of $\sigma^2(R)$ is controlled by the small-k behavior of $S(k)$. Thus, suppressing large-scale density fluctuations is equivalent to requiring $S(k)$ to vanish near the origin. A point pattern is therefore called **hyperuniform** if and only if [3]

$$\lim_{k \to 0} S(k) = 0. \tag{4}$$

When this condition holds, the integrand in Eq. (3) is suppressed at small k, and $\sigma^2(R)$ grows more slowly than R.

2.2 The hyperuniformity exponent and three classes

While the condition $S(0) = 0$ tells us that a pattern is hyperuniform, it does not tell us how strongly the fluctuations are suppressed. The rate at which $S(k)$ vanishes near the origin turns out to control the large-R scaling of $\sigma^2(R)$ and defines a hierarchy of hyperuniformity strength. When $S(k)$ vanishes as a power law,

$$S(k) \sim |k|^\alpha \quad \text{as } k \to 0, \tag{5}$$

the exponent $\alpha > 0$ controls the large-R scaling of the number variance. Substituting this power law into Eq. (3) and taking the $R \to \infty$ limit yields three qualitatively different behaviors [3, 2]:

$$\sigma^2(R) \sim \begin{cases} \text{const} & \alpha > 1 \quad \text{(Class I)}, \\ \Lambda_{\text{II}} \ln R & \alpha = 1 \quad \text{(Class II)}, \\ \Lambda_{\text{III}} R^{1-\alpha} & 0 < \alpha < 1 \quad \text{(Class III)}. \end{cases} \tag{6}$$

Class I is the strongest form of hyperuniformity, as the variance remains bounded no matter how large the window gets. This means that at large enough scales, the pattern looks uniformly ordered. All perfect crystals and quasicrystals generated by Pisot substitution rules [1] belong to this class [5]. For Class II hyperuniformity, the variance grows infinitely with R, but only logarithmically. Class III is the weakest form of hyperuniformity, with an unbounded variance that grows more slowly than that of a Poisson distribution.

For quasicrystals, whose structure factor consists of dense Bragg peaks rather than a smooth envelope, fitting a power law directly to $S(k)$ is impractical. Instead, the exponent can be extracted from the integrated spectral intensity [5, 4]

$$Z(k) = 2 \int_0^k S(q)\, dq \sim k^{1+\alpha} \quad \text{as } k \to 0, \tag{7}$$

which smooths over the individual peaks and provides a clean power law.

2.3 The surface-area coefficient $\bar{\Lambda}$

Knowing which class a pattern belongs to does not fully characterize it. Two Class I quasicrystals can both have $\alpha = 3$ and yet differ substantially in how much their density fluctuates inside a window of any given size. To see this difference, we need a quantity that captures the amplitude of the fluctuations, not just their scaling.

For Class I patterns in one dimension, the variance does not simply converge to a constant but typically oscillates about a mean level. This is especially true for quasicrystals with dense Bragg spectra, where the oscillations persist at all scales. In order to extract a well-defined number from this behavior, we define the surface-area coefficient as the running average

$$\bar{\Lambda} \equiv \lim_{L \to \infty} \frac{1}{L} \int_0^L \sigma^2(R)\, dR. \tag{8}$$

This averages out the oscillations and converges to the mean level of the plateau. When the variance does converge to a pointwise limit, $\bar{\Lambda}$ simply equals that limit.

The name "surface-area coefficient" comes from the d-dimensional generalization, where the leading term of the number variance for Class I patterns scales with the surface area of the observation window [3]. In one dimension, the surface of an interval consists of just two endpoints, so the variance approaches a constant, and $\bar{\Lambda}$ is simply the mean level of that constant.

Torquato and Stillinger proved that the one-dimensional integer lattice achieves $\bar{\Lambda} = 1/6$, which is the global minimum among all infinite one-dimensional hyperuniform patterns [3]. For aperiodic systems, however, almost no values are known. The only prior result is $\bar{\Lambda} = 0.20110$ for the Fibonacci chain, computed by Zachary and Torquato [4].

[1] A Pisot number is an algebraic integer greater than 1 whose algebraic conjugates all have absolute value less than 1. The golden ratio $\tau = (1 + \sqrt{5})/2 \approx 1.618$, whose conjugate is $(1 - \sqrt{5})/2 \approx -0.618$, is the simplest example.

The definition of $\bar{\Lambda}$ in Eq. (8) yields a finite value only for Class I patterns, because the variance is bounded. For Classes II and III, the variance grows without bound and $\bar{\Lambda}$ diverges. To rank patterns in these classes, we need analogous coefficients that extract the leading amplitude from the appropriate growth law. For Class II ($\alpha = 1$), the variance grows logarithmically, so we define

$$\Lambda_{\mathrm{II}} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}, \tag{9}$$

which isolates the prefactor of the logarithmic growth.

For Class III ($0 < \alpha < 1$), the variance grows as a power law and we define

$$\Lambda_{\mathrm{III}} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}, \tag{10}$$

which isolates the prefactor of the power-law growth. The procedures used to extract Λ_{II} and Λ_{III} from finite data are described in Sec. 3.2.

2.4 Substitution tilings and the eigenvalue formula

Many of the patterns we study are generated by substitution rules. A one-dimensional substitution tiling starts with a short seed word over a finite alphabet and grows by repeatedly replacing each letter with a fixed string of letters. After many iterations, the result is a long aperiodic sequence. By assigning each letter a physical tile length and placing the tiles end to end, one can produce a one-dimensional point pattern at the tile boundaries.

The growth of the tiling is governed by the substitution matrix M, an $r \times r$ non-negative integer matrix (for r tile types) whose entry M_{ij} counts how many tiles of type i are produced when a single tile of type j is substituted [5]. For example, the Fibonacci chain uses two tile types, S (short) and L (long), with the rule $S \to L$, $L \to LS$. Its substitution matrix is

$$M_{\mathrm{Fib}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \tag{11}$$

with eigenvalues $\lambda_1 = \tau = (1+\sqrt{5})/2 \approx 1.618$ (the golden ratio) and $\lambda_2 = -1/\tau \approx -0.618$.

The largest eigenvalue λ_1 is always positive and real for non-negative integer matrices. Every time the substitution rule is applied, the total length of the tiling is multiplied by approximately λ_1. The corresponding eigenvector then gives the asymptotic relative frequencies of each tile type, and the tile lengths are set so that the pattern scales consistently. For two tile types, this fixes the length ratio $L/S = \lambda_1$ [5].

Substitution matrices and hyperuniformity are connected by the eigenvalue formula of Oğuz et al [6, 5]:

$$\alpha = 1 - \frac{2\ln|\lambda_2|}{\ln \lambda_1}. \tag{12}$$

For substitution tilings, this equation determines α entirely from the two largest eigenvalues. The formula applies when λ_1 is a Pisot number, meaning all other eigenvalues have absolute values less than 1 [5]. Physically, λ_1 sets how much the tiling grows with each substitution step, while $|\lambda_2|$ controls how fast density fluctuations decay. A smaller $|\lambda_2|$ corresponds to stronger suppression of large-scale density fluctuations and a larger α. From Eq. (12), the tiling is hyperuniform ($\alpha > 0$) whenever $|\lambda_2| < \sqrt{\lambda_1}$.

For any $r \times r$ Pisot substitution matrix M with $|\det M| = 1$, the product of all eigenvalues has absolute value 1 ($|\det M| = |\lambda_1 \cdot \lambda_2 \cdots \lambda_r| = 1$). Then, because $|\lambda_2| \geq |\lambda_j|$ for all $j \geq 2$, it

follows that $\lambda_1 \cdot |\lambda_2|^{r-1} \geq 1$. Because the Pisot condition requires $|\lambda_j| < 1$ for all $j \geq 2$, the second eigenvalue therefore satisfies

$$|\lambda_2| \geq \lambda_1^{-1/(r-1)}. \tag{13}$$

Substituting into Eq. (12), we get an upper bound on α:

$$\alpha \leq 1 + \frac{2}{r-1}. \tag{14}$$

For rank $r = 2$ this gives $\alpha \leq 3$. We can see this in practice by considering the metallic-mean family, with substitution rules $S \to L$, $L \to L^n S$ for integer $n \geq 1$. Each metallic ratio has matrix $M = \left(\begin{smallmatrix} 0 & 1 \\ 1 & n \end{smallmatrix}\right)$ with $\det M = -1$ and eigenvalues $\lambda_1 = \mu_n = \frac{1}{2}(n + \sqrt{n^2+4})$ (the nth metallic mean) and $\lambda_2 = -1/\mu_n$. Because $|\lambda_2| = 1/\lambda_1$, the eigenvalue formula gives

$$\alpha = 1 - \frac{2\ln(1/\mu_n)}{\ln \mu_n} = 1 + 2 = 3 \tag{15}$$

for every n. Thus, the Fibonacci ($n = 1$), silver ($n = 2$), bronze ($n = 3$), and any other metallic-mean chain are all Class I with $\alpha = 3$ regardless of n.

2.5 Uniformly randomized lattices

The metallic-mean chains discussed in Sec. 2.4 are all Class I with $\alpha = 3$. To examine Class I behavior at $\alpha = 2$ and to obtain an exact reference value for $\bar{\Lambda}$, we also study the uniformly randomized lattice (URL), which is constructed by taking the integer lattice and independently displacing each site by a uniform random amount on $[-a/2, a/2]$ [7].

Because the displacements are independent and identically distributed, the only correlations between sites come from the underlying lattice. On large scales, any window of half-width $R \gg a$ then contains close to $2\rho R$ points, with fluctuations only from the boundary region of width $\sim a$ where a displaced site may or may not fall inside. These boundary fluctuations do not grow with R, so $\sigma^2(R)$ is bounded and the URL is Class I hyperuniform. The structure factor vanishes quadratically near the origin, $S(k) \sim k^2$, giving $\alpha = 2$ for all $a > 0$ [7].

Klatt, Kim, and Torquato [7] derived an exact expression for $\bar{\Lambda}$ in one dimension:

$$\bar{\Lambda}(a) = \frac{1 + a^2}{6}. \tag{16}$$

At $a = 0$ this reduces to the integer lattice minimum $\bar{\Lambda} = 1/6$. As a increases, so does $\bar{\Lambda}$: larger perturbations produce more variable density fluctuations.

2.6 Stealthy hyperuniform patterns

Substitution tilings achieve hyperuniformity through their algebraic structure, and perturbed lattices inherit it from the underlying crystal. A third mechanism is to engineer hyperuniformity directly in Fourier space. A stealthy hyperuniform pattern satisfies the condition

$$S(k) = 0 \quad \text{for all } |k| < K^*, \tag{17}$$

where K^* is a finite exclusion radius [8]. The structure factor vanishes not just at $k = 0$ but over an entire interval, making these patterns invisible to scattering at long wavelengths.

Stealthy patterns are realized as ground states of a pair potential that penalizes all density fluctuations at wavevectors $|k| < K^*$ [8]. Minimizing the total potential energy forces $S(k)$ to zero for all $|k| < K^*$, which is exactly the stealthiness condition in Eq. (17).

Stealthy patterns are characterized by the parameter

$$\chi = \frac{M(K^*)}{N-1}, \tag{18}$$

where $M(K^*)$ is the number of independently constrained wavevectors $k = 2\pi m/L$ satisfying $0 < |k| \leq K^*$, and $N-1$ is the number of translational degrees of freedom [8]. The ratio χ measures the fraction of Fourier modes that are constrained to zero. Because $S(k) = 0$ over a finite interval rather than vanishing as a power law $S(k) \sim |k|^\alpha$, stealthy patterns do not fit the standard three-class framework and the exponent α is not defined for them. However, $\bar{\Lambda}$ is still well-defined for stealthy patterns and is what we use to compare them.

3 Methods

3.1 Pattern generation

Substitution tilings are generated by iteratively applying the substitution rules to a seed word until the tiling exceeds 10^6 tiles, which is large enough that finite-size effects on $\sigma^2(R)$ are negligible. Tile lengths are set as described in Sec. 2.4, and points are then placed at tile boundaries to produce the one-dimensional point pattern. The URL patterns are generated by starting from an integer lattice of $N = 50{,}000$ points and displacing each site independently by a uniform random variable on $[-a/2, a/2]$, with a taking 17 values between 0.05 and 2.0 to span the range from weak to strong perturbations. Data for stealthy configurations was provided by a collaborator, with $N = 2{,}000$ points at density $\rho = 1$. For each of three values $\chi \in \{0.1, 0.2, 0.3\}$, we worked with an ensemble of several thousand independent realizations of the stealthy ground state, over which $\sigma^2(R)$ is averaged to obtain a representative ensemble estimate.

3.2 Extraction of α and $\bar{\Lambda}$

We extract two quantities from each pattern: the surface-area coefficient $\bar{\Lambda}$ (or its Class II/III generalization), obtained from the number variance $\sigma^2(R)$ in real space, and the hyperuniformity exponent α, obtained from the eigenvalue formula for substitution tilings (Eq. (12)), analytically for URL patterns [7], and numerically via the spreadability method as an independent check [9].

We compute $\sigma^2(R)$ by randomly placing 30,000 windows of half-width R along the pattern and recording the variance of the point counts across all window placements. We impose periodic boundary conditions on the pattern and restrict window half-widths to $R < L/4$ so that each window samples only a fraction of the total pattern. We evaluate $\sigma^2(R)$ at several thousand values of R, ranging from below the mean point spacing up to this maximum, to resolve the full large-R behavior. For stealthy patterns, where the ensemble captures configuration-to-configuration variability, $\sigma^2(R)$ is computed for each configuration individually and then averaged across the full ensemble before extracting $\bar{\Lambda}$.

We extract $\bar{\Lambda}$ as the running average of $\sigma^2(R)$ over the upper two-thirds of the sampled R range, where the variance has settled into its large-R asymptotic behavior. For quasicrystals whose variance oscillates persistently rather than converging pointwise, this running average converges to a stable mean level, and we take that level as $\bar{\Lambda}$.

For Class II and Class III patterns, $\bar{\Lambda}$ diverges and we instead extract Λ_{II} and Λ_{III} by fitting the large-R variance data to the appropriate scaling forms. For Class II we fit $\sigma^2(R) = \Lambda_{\mathrm{II}} \ln R + b$, where the additive constant b absorbs subleading contributions that would otherwise bias the extracted Λ_{II}. For Class III we fit $\sigma^2(R) = A\,R^\beta$ with both A and β as free parameters, from which $\Lambda_{\mathrm{III}} = A$

and $\alpha = 1 - \beta$. Leaving β free means the fit independently determines α from the variance data alone, instead of just assuming the prediction from Eq. (12).

As an independent numerical check on α, we use the diffusion spreadability method [9, 10, 11]. The spreadability measures how quickly an initially localized concentration field spreads through a heterogeneous medium. For a hyperuniform medium, the long-time decay rate of this spreading is controlled by α, so we can fit the decay rate to get α [9].

To apply the method, each point pattern is embedded in a one-dimensional two-phase medium by centering a solid rod of half-length $a = \phi_2/(2\rho)$ at each point, giving packing fraction $\phi_2 = 0.35$ [11]. From the structure factor $S(k)$, computed via discrete Fourier transform, we obtain the spectral density [9]

$$\tilde{\chi}_V(k) = \rho\, |\tilde{m}(k)|^2\, S(k), \tag{19}$$

where $\tilde{m}(k)$ is the Fourier transform of the rod indicator function. The excess spreadability is then [9, 10]

$$\mathcal{S}(t) = \frac{d_k}{\pi\phi_2} \sum_n \tilde{\chi}_V(k_n)\, e^{-k_n^2 Dt}, \tag{20}$$

where $d_k = 2\pi/L$ is the wavevector spacing and D is the diffusion coefficient. For a Class I hyperuniform medium, $\mathcal{S}(t)$ decays as a power law at long times [9]:

$$\mathcal{S}(t) \sim t^{-(1+\alpha)/2}, \tag{21}$$

and fitting $\ln \mathcal{S}$ vs. $\ln t$ yields α. We use spreadability as a numerical check on α for all substitution tilings in this study.

3.3 Validation

We verify the variance estimation procedure against three benchmarks of increasing complexity before applying it to the systems of interest. First, applying the procedure to a Poisson process recovers the theoretical linear scaling $\sigma^2(R) = 2\rho R$ to within 1.1%, confirming that the sliding-window algorithm is unbiased. Second, the URL exact formula $\bar{\Lambda} = (1 + a^2)/6$ is reproduced to better than 0.3% across all 7 values of a tested, verifying that the running-average extraction works correctly for a correlated system with known $\bar{\Lambda}$. Third, our computed value $\bar{\Lambda} = 0.2000$ for the Fibonacci chain agrees with the literature value $\bar{\Lambda} = 0.20110$ of Zachary and Torquato [4] to within 0.55%, which is the only prior computation of $\bar{\Lambda}$ for a quasicrystal.

4 Results

4.1 Metallic-mean family

We begin with the metallic-mean family $S \to L$, $L \to L^n S$ for $n = 1$ through 5, which spans the Fibonacci, silver, bronze, copper, and nickel chains (Fig. 2). As n grows, the long tile becomes much longer than the short tile: short tiles make up 36% of the Fibonacci sequence but only 8% at $n = 10$.

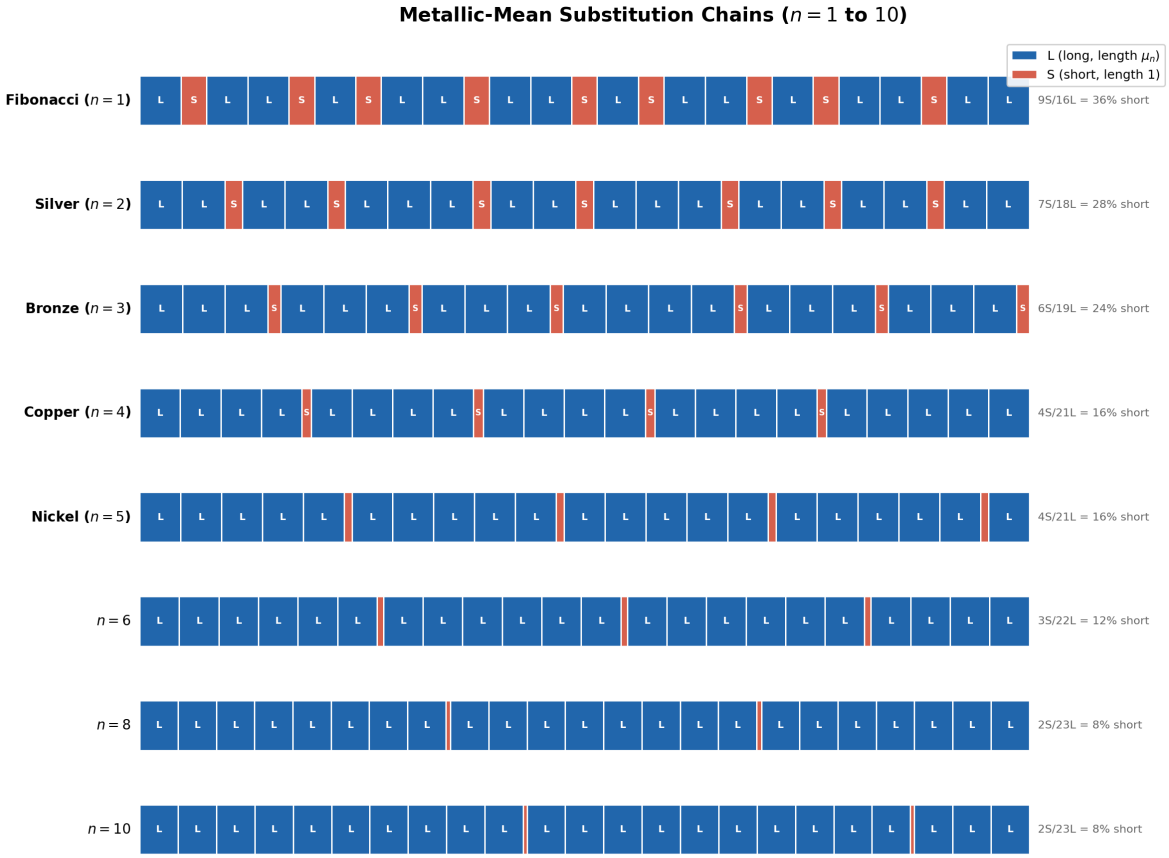

Figure 2: Metallic-mean tile sequences for $n = 1$ (Fibonacci) through $n = 10$, 25 tiles each. Long tiles (blue) have length μ_n; short tiles (red) have length 1.

As shown in Sec. 2.4, the eigenvalue formula predicts $\alpha = 3$ for every member of this family regardless of n. Table 1 confirms this: the spreadability-based estimates of α agree with the prediction to within 4% for all five chains, with accuracy decreasing at larger n due to stronger log-periodic oscillations in $\mathcal{S}(t)$. The integer lattice ($\bar{\Lambda} = 1/6$) is included as the proven global minimum [3]; only the Fibonacci $\bar{\Lambda}$ was previously known [4].

n	Chain	μ_n	ρ	α (eigenvalue)	α (spreadability)	$\bar{\Lambda}$
—	Integer lattice	—	1.000	∞	—	0.167
1	Fibonacci	1.618	0.724	3	3.05	0.200
2	Silver	2.414	0.500	3	2.99	0.250
3	Bronze	3.303	0.361	3	2.99	0.282
4	Copper	4.236	0.276	3	3.03	0.293
5	Nickel	5.193	0.222	3	2.88	0.310

Table 1: Metallic-mean chains for $n = 1$ through 5.

All five chains have $\alpha = 3$, despite having very different tile-length ratios and overall structures. This demonstrates that α is determined entirely by the eigenvalues of the substitution matrix, not

by the geometry of the tiles. For each pattern, the variance is bounded and oscillates about a mean level $\bar{\Lambda}$.[2] These $\bar{\Lambda}$ values are all different, increasing monotonically from 0.200 (Fibonacci) to 0.310 (Nickel), which shows that α and $\bar{\Lambda}$ indeed capture independent aspects of the pattern. While α controls how fast $S(k)$ vanishes near the origin, $\bar{\Lambda}$ measures the amplitude of the remaining density fluctuations.

4.2 Structural constraints on α

The bound $\alpha \leq 1 + 2/(r-1)$ from Eq. (14) locks rank-2 matrices with $|\det M| = 1$ at $\alpha = 3$, as confirmed in Sec. 4.1. To reach other values of α, we can either increase the rank of the substitution matrix or relax the condition $|\det M| = 1$.

Increasing to rank 3 lowers the bound to $\alpha \leq 2$ and opens the range $1 < \alpha \leq 2$. The Bombieri-Taylor substitution [12] is a well-known example: its three-letter rule $a \to aac$, $b \to ac$, $c \to b$ has substitution matrix

$$M_{\mathrm{BT}} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

with $|\det M| = 1$ and largest eigenvalue $\lambda_1 \approx 2.247$. The eigenvalue formula gives $\alpha = 1.545$, and we measure $\bar{\Lambda} = 0.377$.

Among rank-3 matrices, a special case arises when λ_2 and λ_3 form a complex conjugate pair. The condition $|\det M| = \lambda_1 |\lambda_2|^2 = 1$ then forces $|\lambda_2| = 1/\sqrt{\lambda_1}$, and substituting into the eigenvalue formula gives $\alpha = 2$ exactly, regardless of λ_1. We verified this by exhaustively searching all 39,304 rank-3 matrices with non-negative integer entries and row sums ≤ 4: of these, 504 are Pisot matrices with a complex eigenvalue pair, spanning 18 distinct values of λ_1 from 1.32 to 3.85, and all give $\alpha = 2$. One example is the rule $a \to bc$, $b \to c$, $c \to abc$, which gives $\bar{\Lambda} = 0.275$.

Together, these results show that no substitution matrix with $|\det M| = 1$ can produce α between 2 and 3: rank 2 is locked at $\alpha = 3$, while rank ≥ 3 gives $\alpha \leq 2$. A further search over 24 million rank-4 matrices (row sums ≤ 3) confirmed this gap.

However, this gap can be filled by allowing $|\det M| > 1$. For a rank-2 matrix, $|\det M| = \lambda_1 |\lambda_2|$, so $|\lambda_2| = |\det M|/\lambda_1$. Substituting into Eq. (12) gives

$$\alpha = 3 - \frac{2 \ln |\det M|}{\ln \lambda_1}, \tag{22}$$

which reduces to $\alpha = 3$ when $|\det M| = 1$ but yields $\alpha < 3$ when $|\det M| > 1$. For α to remain above 2, we need $|\det M| < \sqrt{\lambda_1}$. Since $|\det M|$ is a positive integer, the smallest useful value is $|\det M| = 2$, which requires $\lambda_1 > 4$. The simplest such matrix is $M = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 4 \end{smallmatrix}\right)$ with $\lambda_1 = 2 + \sqrt{6} \approx 4.449$, giving $\alpha = 2.071$. A systematic search over all rank-2 matrices with row sums ≤ 10 found 138 matrices with $|\det M| = 2$ and α between 2.071 and 2.393, spanning 12 distinct α values. Table 2 shows the 7 most populated α values. Within each group, $\bar{\Lambda}$ varies by factors of 5–15, showing that even matrices with identical eigenvalues can produce very different levels of density fluctuation.

[2] Plots of $\sigma^2(R)$ for the integer lattice and three metallic-mean quasicrystals are shown in Appendix **??**.

α	Chains	$\bar{\Lambda}$ range
2.071	12	[0.32, 2.01]
2.087	10	[0.34, 1.57]
2.175	18	[0.32, 3.36]
2.271	16	[0.34, 3.76]
2.301	18	[0.33, 4.05]
2.362	12	[0.34, 4.80]
2.393	4	[0.34, 5.08]

Table 2: Rank-2 matrices with $|\det M| = 2$ and $\alpha \in (2, 3)$. 138 chains total across 12 distinct α values; the 7 most populated are shown.

4.3 Classes II and III

The patterns discussed so far are all Class I ($\alpha > 1$). To complete the picture, we also examine two substitution tilings that fall into the weaker hyperuniformity classes.

The period-doubling chain, with rule $S \to L$, $L \to LSS$ and matrix $M = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, has eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. Because $|\lambda_2| = 1$, the eigenvalue formula gives $\alpha = 1$ exactly, placing it at the boundary between Class I and Class II. Its variance grows logarithmically, and fitting $\sigma^2(R) = \Lambda_{\mathrm{II}} \ln R + b$ yields $\Lambda_{\mathrm{II}} = 0.080$.

The 0222 chain, with rule $S \to LL$, $L \to SSLL$ and matrix $M = \left(\begin{smallmatrix} 0 & 2 \\ 2 & 2 \end{smallmatrix}\right)$, has $\lambda_1 = 1+\sqrt{5} \approx 3.236$ and $|\lambda_2| = \sqrt{5} - 1 \approx 1.236$. Since $|\lambda_2| > 1$, this is not a Pisot substitution, and the eigenvalue formula gives $\alpha = 0.639$, placing it in Class III. Its variance grows as a power law, and the free-parameter fit $\sigma^2(R) = A\,R^{\beta}$ yields $\Lambda_{\mathrm{III}} = 0.209$ with $\alpha = 1 - \beta = 0.68$. This is within 6% of the eigenvalue prediction $\alpha = 0.639$; the discrepancy reflects the difficulty of extracting a clean power-law exponent from finite variance data with oscillations.

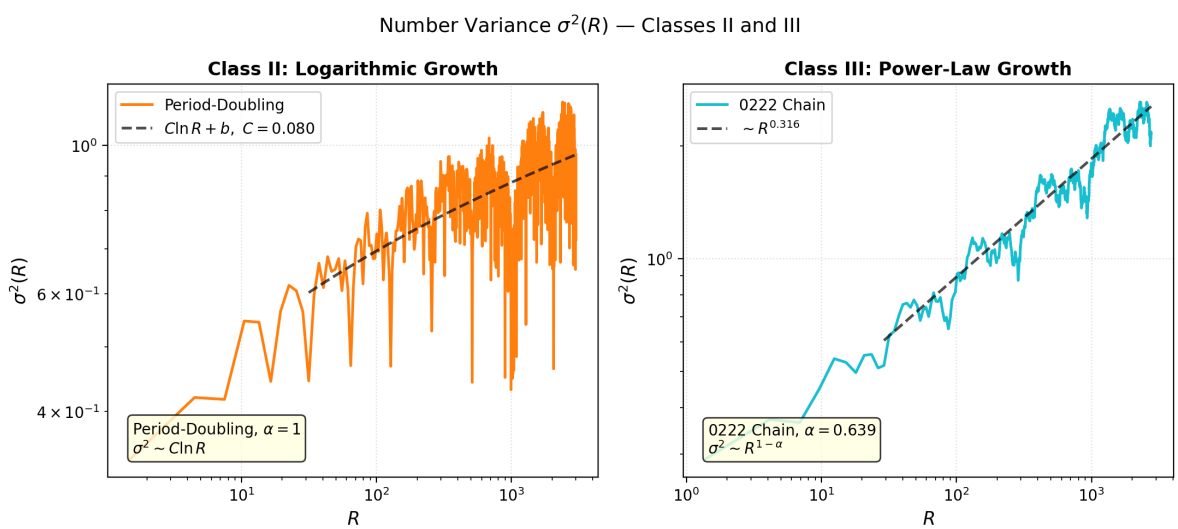

Figure 3: Number variance $\sigma^2(R)$ for Classes II and III. Left: Period-Doubling (Class II, $\alpha = 1$), with logarithmic growth and fitted coefficient $\Lambda_{\mathrm{II}} = 0.080$. Right: 0222 chain (Class III, $\alpha = 0.639$), with power-law growth $\sigma^2 \sim R^{0.316}$. Dashed lines show the fits described in Sec. 3.2.

4.4 Ranking Class I patterns by $\bar{\Lambda}$

We now turn to ranking patterns by $\bar{\Lambda}$.

As shown in Table 1 and Fig. 2, $\bar{\Lambda}$ increases with n as the tile-length disparity grows. Extending the computation to larger n reveals that $\bar{\Lambda}$ appears to approach a limiting value near 1/3 as $n \to \infty$ (Fig. 4). Whether this limit is exactly 1/3 remains an open question.

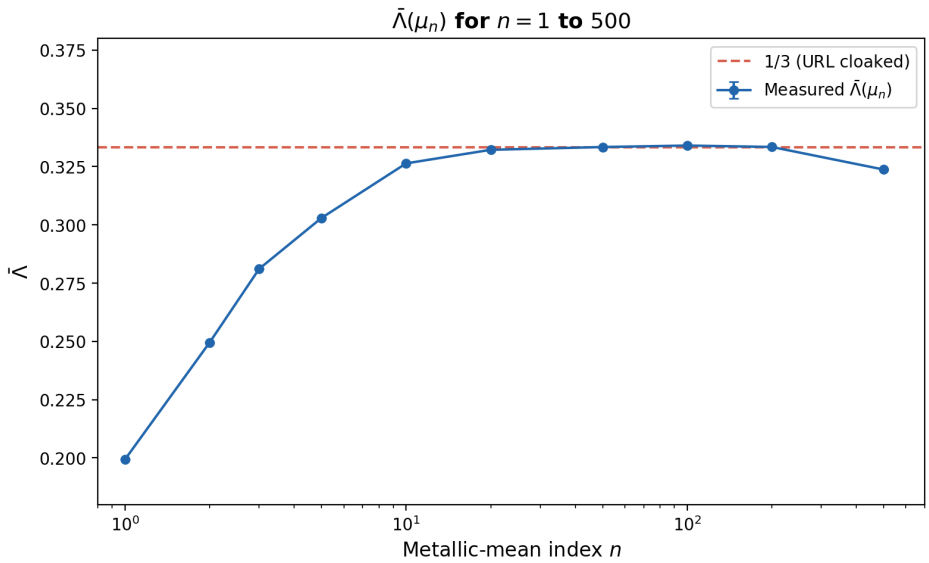

Figure 4: $\bar{\Lambda}$ for the metallic-mean family, computed for $n = 1$ through 500. The dashed line marks 1/3 as a visual reference. Error bars grow at large n due to the coarse quasi-periodic structure of high-n chains.

The URL provides an exact benchmark: $\bar{\Lambda}(a) = (1 + a^2)/6$ [7]. Our numerical measurements track this formula to better than 0.3% across all 17 values of a tested (Sec. 3.3).

For stealthy configurations, $\bar{\Lambda}$ decreases as χ increases: $\bar{\Lambda} = 1.021$ at $\chi = 0.1$, 0.526 at $\chi = 0.2$, and 0.357 at $\chi = 0.3$. Constraining more Fourier modes to zero leaves less room for density fluctuations, reducing $\bar{\Lambda}$ toward the lattice minimum.

Table 3 collects the full catalog of patterns studied in this work, sorted by $\bar{\Lambda}$. The metallic-mean chains all share $\alpha = 3$ but span a factor of 1.5 in $\bar{\Lambda}$, and the 138 matrices with $|\det M| > 1$ (Table 2) show even larger variation within each α group. Knowing α alone is not enough to predict $\bar{\Lambda}$.

Pattern	Class	α	$\bar{\Lambda}$
Integer lattice	I	∞	$1/6 = 0.167$
Fibonacci ($n = 1$)	I	3	0.201
Silver ($n = 2$)	I	3	0.250
Cubic $\alpha = 2$	I	2	0.275
Bronze ($n = 3$)	I	3	0.282
Copper ($n = 4$)	I	3	0.293
Nickel ($n = 5$)	I	3	0.310
URL ($a = 1$)	I	2	$1/3 = 0.333$
Stealthy $\chi = 0.3$	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Stealthy $\chi = 0.2$	I	—	0.526
Stealthy $\chi = 0.1$	I	—	1.021
Period-Doubling	II	1	$\Lambda_{\mathrm{II}} = 0.080$
0222 chain	III	0.639	$\Lambda_{\mathrm{III}} = 0.209$

Table 3: Full catalog of patterns studied in this work, sorted by $\bar{\Lambda}$. The integer lattice achieves the proven global minimum $\bar{\Lambda} = 1/6$. Stealthy patterns do not have a power-law α. Period-Doubling and 0222 chain have divergent $\bar{\Lambda}$ and are characterized by Λ_{II} and Λ_{III} instead.

5 Conclusion

I am still compiling and organizing my results for this report. I am thinking of additionally including are the Yuan-Torquato (2026) improved spreadability method (and related analysis on why the original spreadability converged so slowly for the Bombieri-Taylor chain), in addition to continuing to develop the theory and results sections.

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A Additional figures

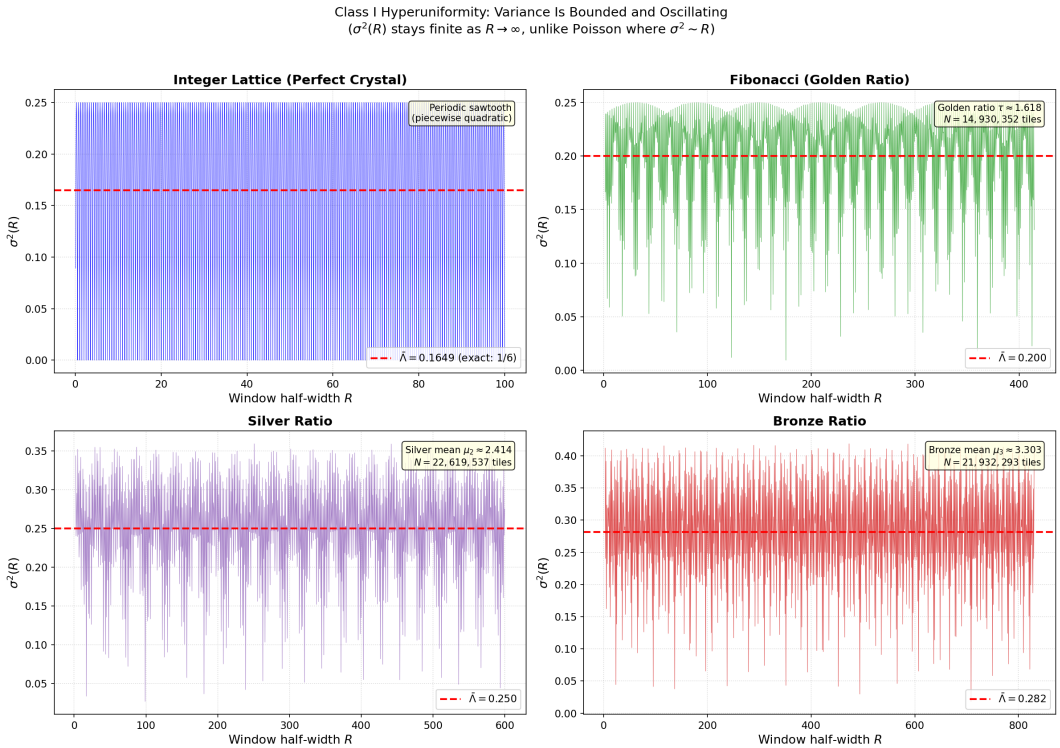

Figure 5: Bounded number variance for the integer lattice and three metallic-mean chains (all Class I). Dashed lines mark $\bar{\Lambda}$.

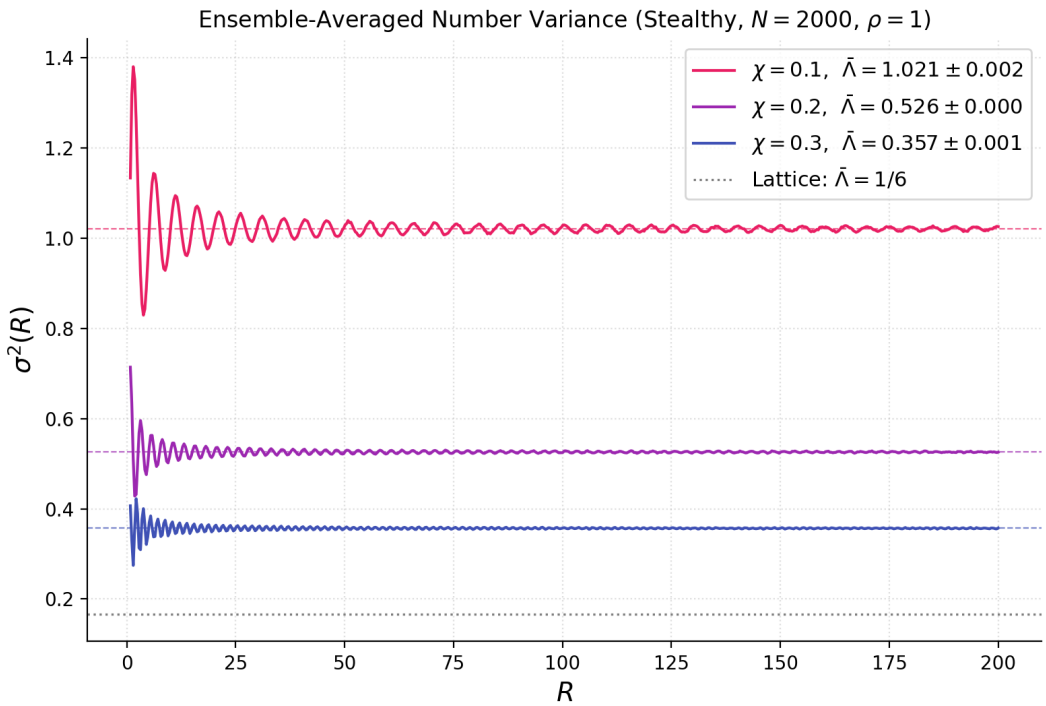

Figure 6: Ensemble-averaged number variance for stealthy configurations at $\chi = 0.1$, 0.2, and 0.3. $\bar{\Lambda}$ decreases with increasing χ; the lattice minimum 1/6 is shown for reference.

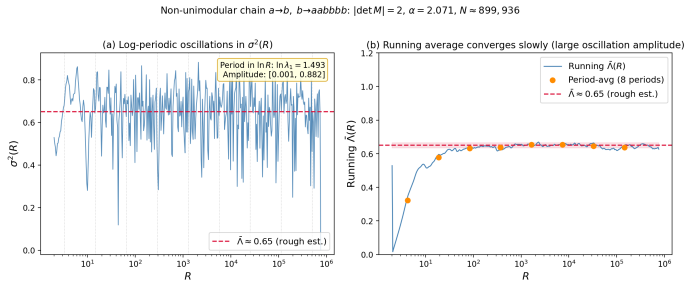

Figure 7: Number variance and running $\bar{\Lambda}(R)$ for the simplest non-unimodular chain ($|\det M| = 2$, $\alpha = 2.071$). Large log-periodic oscillations make $\bar{\Lambda}$ difficult to extract precisely.

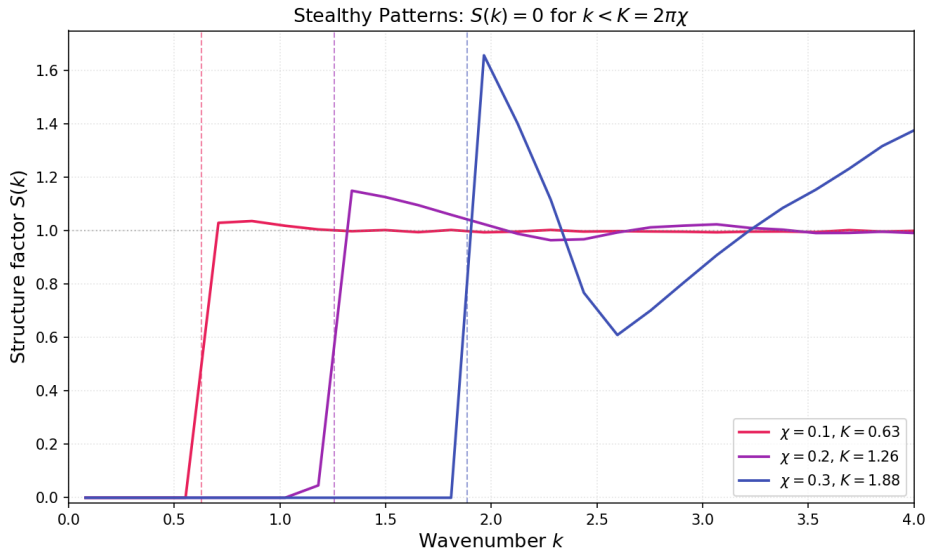

Figure 8: Structure factor $S(k)$ for stealthy configurations at three χ values. $S(k) = 0$ for $k < K^* = 2\pi\chi$.

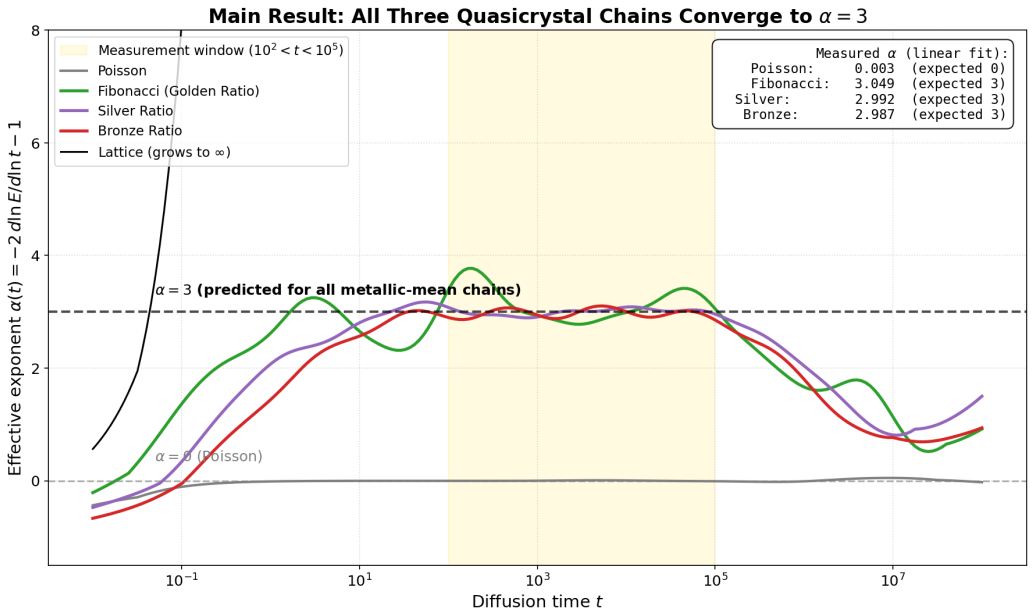

Figure 9: Effective exponent $\alpha(t)$ from spreadability for three metallic-mean chains and a Poisson control. All three chains converge to $\alpha = 3$.

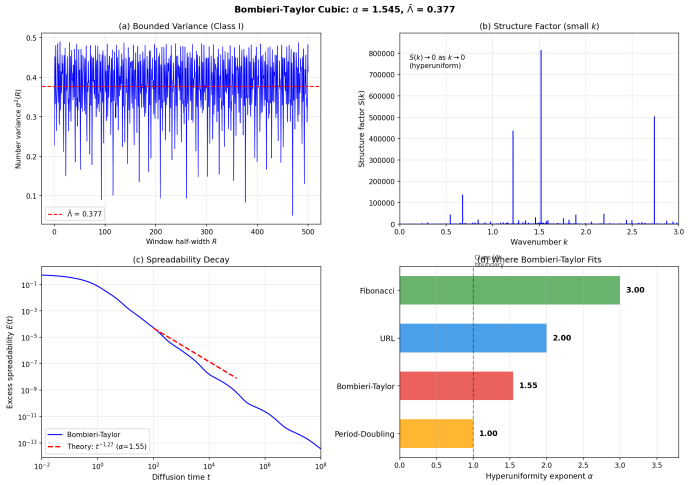

Figure 10: Bombieri-Taylor chain ($\alpha = 1.545$): variance, structure factor, spreadability decay, and placement among known α values.

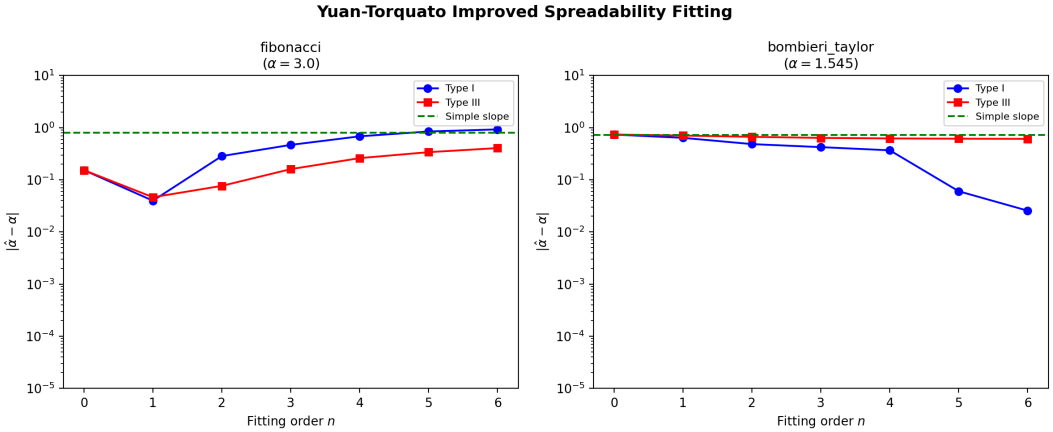

Figure 11: Yuan-Torquato improved spreadability fitting: absolute error $|\hat{\alpha} - \alpha|$ vs. fitting order for Fibonacci and Bombieri-Taylor. Type I order 1 gives 1.3% error for Fibonacci; order 6 gives 1.7% for BT.

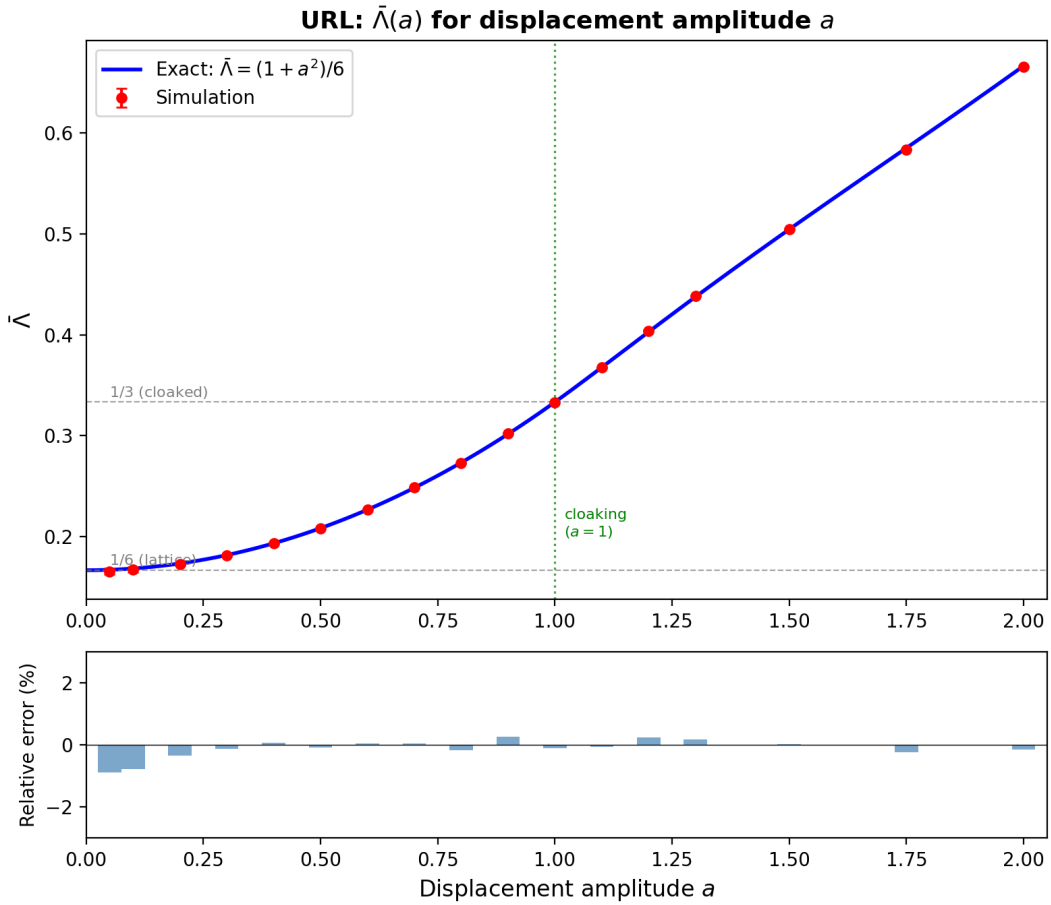

Figure 12: URL validation: numerical $\bar{\Lambda}(a)$ vs. the exact formula $(1 + a^2)/6$ for 17 values of a. Residuals are below 0.3% throughout.

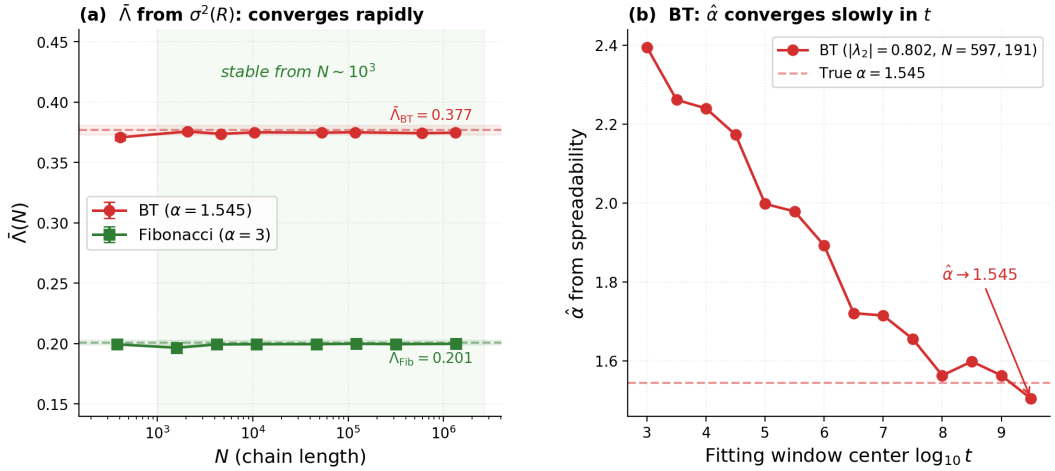

Figure 13: Finite-size convergence. (a) $\bar{\Lambda}$ from $\sigma^2(R)$ stabilizes by $N \sim 10^3$. (b) BT spreadability $\hat{\alpha}$ converges slowly in the fitting window, reaching the true value only at large t.

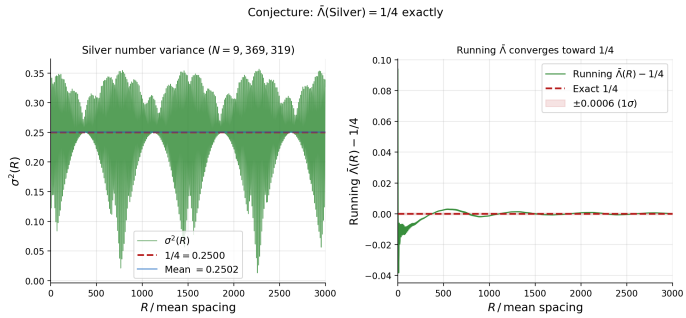

Figure 14: Silver chain number variance ($N = 9.4 \times 10^6$). The running mean $\bar{\Lambda} = 0.2501 \pm 0.0004$ is within 0.3σ of $1/4$.

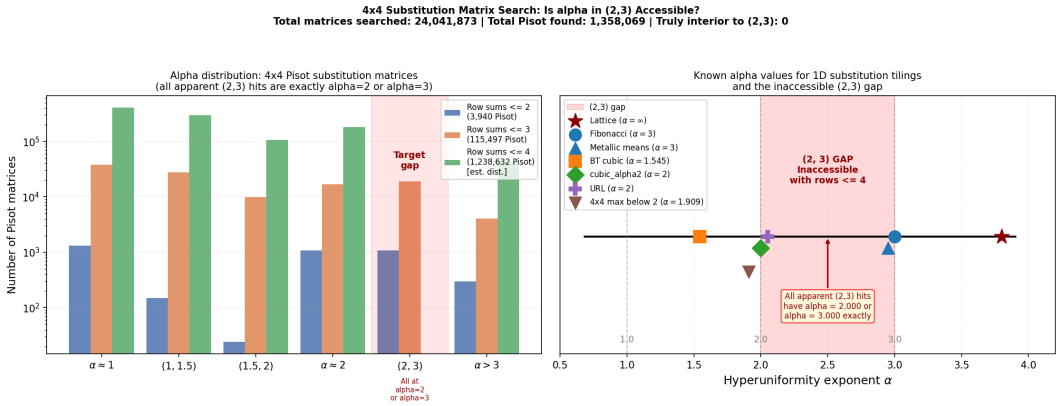

Figure 15: Distribution of α values from the 4×4 substitution matrix search (24M matrices). No unimodular matrix produces $\alpha \in (2, 3)$; all apparent hits land at exactly 2 or 3.

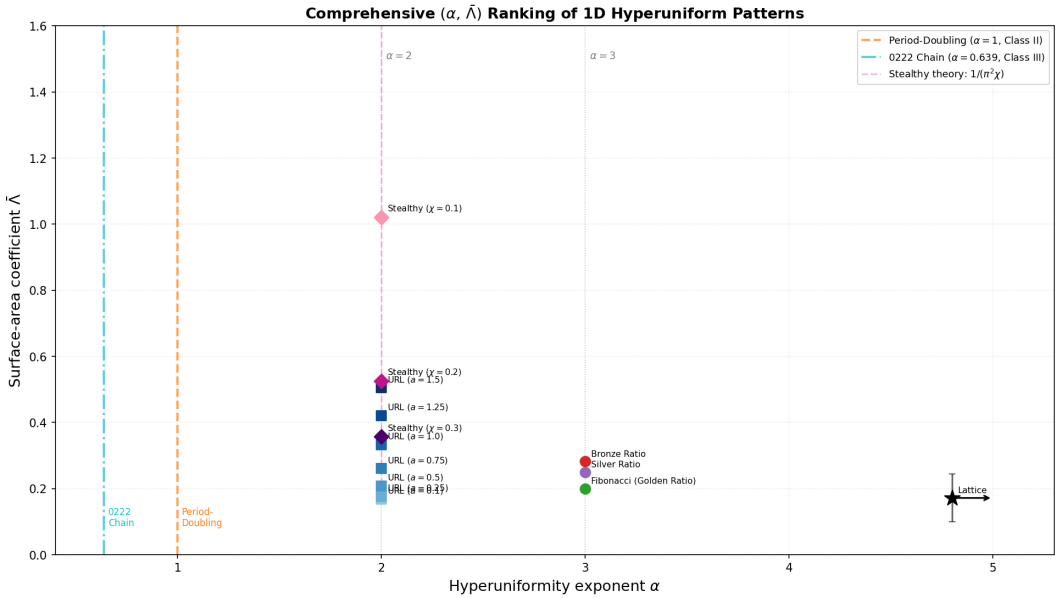

Figure 16: All patterns studied in this work, plotted in the $(\alpha, \bar{\Lambda})$ plane. The lattice sits at the lower-right corner; Class II and III patterns are marked as vertical lines at their respective α.